Sahil Betal

+91-9096188127 betal.sahil13@gmail.com linkedin.com/in/sahil-betal github.com/Lucifer267 Pune, India

SUMMARY

Backend-focused Computer Science student experienced in building scalable APIs, distributed data systems, and security research tools. Skilled in Python, FastAPI, Flask, and cloud-native backend development with emphasis on performance, observability, and system reliability.

EDUCATION

Vishwakarma Institute of Technology *Aug 2023 – May 2027*
Bachelor of Technology in Computer Science *Pune, India*

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript
Backend Development: FastAPI, Flask, REST APIs, Node.js, Async Programming, API Design
Systems & DevOps: Linux, Docker, Kubernetes, Nginx, Git, CI/CD, Shell Scripting
Cloud & Data: AWS, GCP (Vertex AI, Cloud Run), MySQL, Pandas, NumPy, Plotly, Matplotlib, ETL Pipelines

SOFTWARE ENGINEERING PROJECTS

Hybrid Graph RAG Medical QA System *2025*
Python, Flask, Neo4j, Chroma, LLM Systems

- Engineered a hybrid Retrieval-Augmented Generation architecture combining vector search and knowledge graph retrieval for medical question answering.
- Integrated FDA DailyMed drug labels and MedlinePlus datasets to build a structured medical knowledge base.
- Implemented parallel QA pipelines (LLM-only, Vector RAG, Hybrid Graph RAG) with automated evaluation to select the most accurate response.
- Built a Flask backend exposing REST APIs and an interactive dashboard for query comparison and performance metrics.

Custom Honeypot for Multi-Protocol Threat Analysis *2025*
Python, FastAPI, Networking, Systems Engineering

- Developed a lightweight multi-protocol honeypot emulating HTTP, SSH, FTP, and SMB services to monitor real-world malicious traffic.
- Captured and analyzed 150+ intrusion attempts originating from 30+ attacker IPs for behavioral threat analysis.
- Designed an event-driven logging pipeline and asynchronous network handlers to handle concurrent connections reliably.

AURA — Data Visualization & Analytics Platform *2025*
Flask, Plotly, Backend Systems

- Developed a backend analytics platform enabling automated ingestion, statistical analysis, and visualization of CSV datasets.
- Implemented REST APIs for dataset processing and dynamic chart generation used for exploratory data analysis workflows.
- Improved dashboard responsiveness through batching and caching within the data processing pipeline.

MediaPipe Face Landmarker *2024*
JavaScript, MediaPipe, Computer Vision

- Built a real-time facial landmark detection system enabling avatar animation using MediaPipe.
- Integrated 52 facial blendshape parameters for accurate expression tracking and avatar control.
- Implemented a low-latency rendering pipeline supporting 20+ animated avatars simultaneously.

ACHIEVEMENTS & RESEARCH

Presented *Facial Landmark Detection and Avatar Animation* at ICAIN 2025 (Springer Proceedings).

Research work *Beta-Carotene: Molecular Structure C40H56* currently under review in IJSRR (ISSN: 2279-543X).

Ranked 13th among 110+ teams at InnoHack for backend architecture of CodeDebugger & Run.

CERTIFICATIONS

Cloud Technology Consultant — Coursera

AWS Cloud Solutions Architect — Coursera

NVIDIA Deep Learning Certificate — NVIDIA

Machine Learning Operations (MLOps) for Generative AI — Google

Machine Learning Operations (MLOps) with Vertex AI: Model Evaluation — Google